# Pending work and what the evidence supports

## Confirmation: training complete, evaluation pending

Six final step2000 checkpoints exist: MSE and locked AMSE recipe at seeds 22, 23 and 24. Evaluation uses 64 seasonal 2023 starts and averages separate-model metrics, never forecast fields. The recipe is lambda=0.25 with precipitation lambda=0; it was not qualified on development. Selection is locked before confirmation.

## MGDA: blocked at engineering validation

Calibration job 15069232 completed. Smoke jobs 15069233_0 and _1 reached 100 updates but failed uninterrupted-versus-resumed bitwise residual-parameter equality. A failure shared by MGDA and MSE does not demonstrate failure of the scientific objective. No production forecast comparison is available.

## 2023 scope

The eight-case spectral diagnostics have completed and were inspected during development. The separate 122-start year-round memory/accuracy campaign remains pending numerical qualification; the earlier daily campaign failed baseline agreement. The 64-start AMSE confirmation is a third, separate evaluation.

## Supported conclusion

Full AMSE produces materially better power fidelity with worse exact forecast loss. Weak AMSE produces an intermediate trade-off. The 500-update continuations, feedback tests and adaptive variants have not established a joint improvement. Small differences are particularly sensitive to observed numerical drift.

## Recommended next evidence

Finish the already queued confirmation evaluations and inspect all fields, all seeds and paired intervals.
Repair and qualify resume/numerical behavior before running MGDA production or releasing year-round evaluation.
Preserve negative outcomes and keep selection fixed.